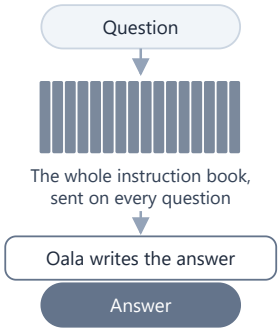
Three ways to run Oala, compared

Vanilla (AI Studio) · EXP-RAG · Multi-Agent · Yuzee AI Token Lab · test run 5 October 2026

We asked all three modes the same 11 questions. Two independent AI judges ranked the answers without knowing which mode wrote which. **On quality the three modes were effectively tied.** Their average places were 1.82 for Vanilla, 1.91 for Multi-Agent and 1.95 for EXP-RAG (out of 3), a gap too small to call on 11 questions. **EXP-RAG** was the fastest, with a typical answer finished in 14.8 s (16.6 s for Multi-Agent and 17.6 s for Vanilla). **EXP-RAG** was the cheapest at $0.0027 per answer, about a quarter of a US cent, 31% less than Vanilla; Multi-Agent cost $0.0033 and Vanilla cost $0.0039. Overall, EXP-RAG led on speed and cost.

How each mode works

Vanilla (AI Studio)



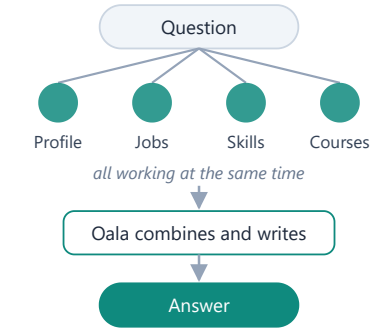
Oala gets the full instruction book on every question: about 46,000 tokens (a token is roughly three-quarters of a word). It also gets facts looked up from our data warehouse. One model call writes the answer, then a review step checks it. Simple and complete, but the model reads a lot of text it does not need.

EXP-RAG



Oala keeps a short core set of rules that is stored and reused, so it is cheap to resend. For each question it adds only the guidance, warehouse facts and memory of the chat that this question needs. The model reads far less text for the same job.

Multi-Agent



EXP-RAG plus a team of small, fast helper models. Each helper has one narrow job (profile, jobs, skills, market, courses, pathway, company or opportunities) and searches the warehouse for itself. They work at the same time. Their findings are merged and checked against the evidence, and Oala writes one answer from them. Simple questions skip the team.

Quality

Each answer was judged blind: two independent AI judges saw the answers shuffled and with the mode names hidden. For every question they put the answers in order from best to worst and scored each one from 1 to 5 on five points.

1.82
Vanilla (AI Studio): average place (1 = best, 3 = worst). First place in 10 of 22 judgements.

1.95
EXP-RAG: average place (1 = best, 3 = worst). First place in 7 of 22 judgements.

1.91
Multi-Agent: average place (1 = best, 3 = worst). First place in 7 of 22 judgements.

Average score (1 to 5)	Vanilla	EXP-RAG	Multi-Agent
Answers the question (relevance)	4.45	4.45	4.45
Backed by real data (grounding)	4.77	4.50	4.32
Fits the person (personalisation)	3.41	3.59	3.45
Clear next steps (actionability)	3.73	3.73	3.59
Easy to read (clarity)	4.14	4.05	4.27
Total (out of 25)	20.5	20.3	20.1

Bars start at the lowest possible score (1, or 5 for the total), so small differences are visible. The two judges picked the same best answer on 9 of 11 questions.

Team questions only. On the 7 questions where Multi-Agent used its team, we also tried a fast setting (the writer thinks less). Ranked among four answers there (1 = best, 4 = worst): Vanilla (AI Studio) 1.86, Multi-Agent 2.14, EXP-RAG 2.21 and Multi-Agent · fast setting 3.29.

Speed

17.6 s
Vanilla (AI Studio): typical time to a finished answer. First words after 10.6 s. Fastest on 0 of 11 questions.

14.8 s
EXP-RAG: typical time to a finished answer. First words after 8.7 s. Fastest on 8 of 11 questions.

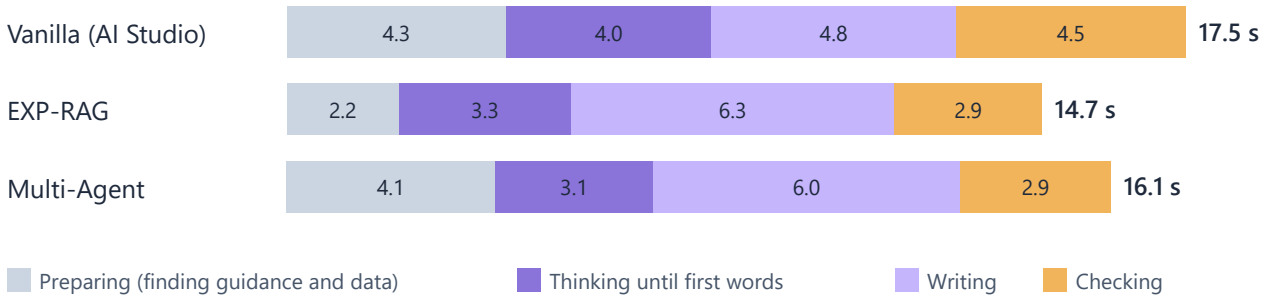
16.6 s
Multi-Agent: typical time to a finished answer. First words after 9.5 s. Fastest on 3 of 11 questions.

"Typical" is the median: half the answers were quicker, half were slower.

Typical (median) time	All 11 questions		Team questions only (7)	
	Finished answer	First words	Finished answer	First words
Vanilla (AI Studio)	17.6 s	10.6 s	15.1 s	7.5 s
EXP-RAG	14.8 s	8.7 s	14.6 s	8.7 s
Multi-Agent	16.6 s	9.5 s	17.6 s	11.5 s
Multi-Agent · fast setting	–	–	13.7 s	6.7 s

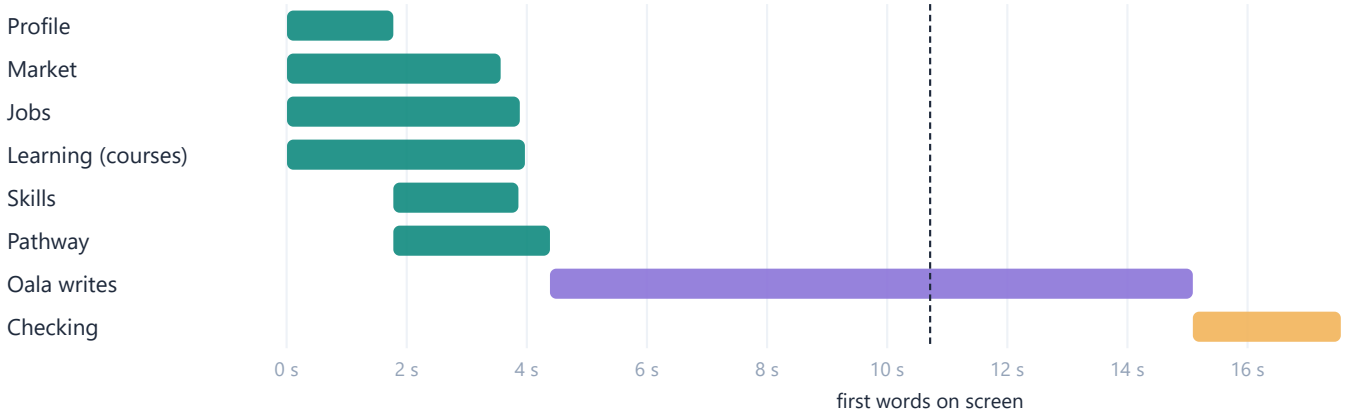
The fast setting was only run on the team questions, so it has no all-question figures and is left out of the summary above.

Where the time goes (average seconds per answer)



Inside one Multi-Agent answer: the team works side by side

Question: "*What could I move into with regular business hours, and what would I need to study?*". Each bar is one helper. Bars that overlap ran at the same time. Then Oala writes the answer and it is checked.



Multi-Agent ran a team of helpers on 7 questions, one quick lookup on 2, and skipped the team on 2. While the team works, the screen shows live progress, so the wait before the first words is not a blank screen.

Why isn't Multi-Agent much faster?

The helpers do work side by side, as the timeline above shows. But the final answer is still written by one model after they finish, and that writer's careful thinking (a few seconds) is what makes the answers good. Working in parallel can shorten the gathering, not the writing. We cut the team's part from 5–8 s to about 2 s after the data lookup. Turning the writer's thinking down made Multi-Agent the fastest mode on team questions (13.7 s typical), but blind judges ranked those answers last. So the default keeps careful thinking, and "Low" in the thinking menu is the fast option.

Speed vs quality: the three versions we tried

Version	Quality vs the other two modes	Typical time, team questions
First version: Pathway waits for the others, detailed helper notes (round 1)	Best: average place 1.64 vs EXP-RAG 2.05, Vanilla 2.09 (all 11 questions)	18.8 s EXP-RAG 13.7 s, Vanilla 18.8 s
Shipped default: lean helpers all working in parallel, careful writer	Tied: average place 1.91 vs EXP-RAG 1.95, Vanilla 1.82	17.6 s EXP-RAG 14.6 s, Vanilla 15.1 s
Fast setting: writer thinking set to low	Lowest: average place 3.29 among 4	13.7 s

Each round used a different run of the questions, so compare within a row, not across rows.

What we changed to make it faster

- Helpers now answer in a smaller, fixed format, so each one takes about 2 s.
- The Pathway helper works alongside the others instead of waiting for all of them.
- The Market check no longer waits about 2 s for graph and forecast data. Oala still receives that data.
- Answers name their sources, do not push a generic follow-up question, and name local providers when a place is given.
- One kind of broken answer format is now repaired automatically, for every mode.

Cost

$0.0039
Vanilla (AI Studio): average cost per answer, about two-fifths of a US cent. That is about 257 answers per US dollar.

$0.0027
EXP-RAG: average cost per answer, about a quarter of a US cent. That is about 373 answers per US dollar.

$0.0033
Multi-Agent: average cost per answer, about a third of a US cent. That is about 299 answers per US dollar.

Per answer, on average	Vanilla	EXP-RAG	Multi-Agent
Main answer	$0.0036	$0.0025	$0.0026
Helper calls (review, lookups, and the Multi-Agent team)	$0.0003	$0.0001	$0.0008
Total	**$0.0039**	**$0.0027**	**$0.0033**
Text the main model reads (tokens; about ¾ of a word each)	60,914	21,455	22,824

Costs use Google Gemini list prices for the recorded usage. Vanilla is dearer mainly because it resends the whole instruction book every time.

Question by question

Question	Best answer (judges' average)	Vanilla	EXP-RAG	Multi-Agent
What does RPL mean?	EXP-RAG	22.4 s	**13.5 s**	14.8 s no team
How many UX designer jobs are in Perth?	Multi-Agent	17.6 s	15.4 s	**14.1 s** 1 quick lookup
Am I suitable for UX designer jobs? I've been a cafe supervisor for 4 years and I taught myself Figma.	Vanilla	14.5 s	**13.2 s**	16.7 s team of 3 fast setting 13.7 s
I want to move from hospitality into UX design. Give me the fastest realistic path.	EXP-RAG	22.7 s	**14.6 s**	17.8 s team of 6 fast setting 15.7 s
What do I need to improve to get a product role at Canva?	Multi-Agent	13.8 s	**13.1 s**	19.4 s team of 5 fast setting 14.1 s
Find teaching assistant jobs for me in Melbourne.	Multi-Agent	15.1 s	**14.8 s**	15.1 s team of 3 fast setting 13.4 s
What jobs suit me?	Vanilla	13.4 s	**9.0 s**	9.7 s team of 3 fast setting 11.6 s
I'm 45, I've managed retail stores for 20 years and want to move into HR. What are my options?	EXP-RAG	19.2 s	**19.1 s**	21.9 s team of 6 fast setting 17.2 s
What TAFE courses in Geelong could help me become a disability support worker?	Vanilla	15.6 s	**15.3 s**	16.6 s 1 quick lookup
I'm a registered nurse in Brisbane and I'm burnt out from shift work.	EXP-RAG	19.3 s	16.1 s	**13.3 s** no team
(follow-up) What could I move into with regular business hours, and what would I need to study?	Vanilla	19.4 s	18.0 s	**17.6 s** team of 6 fast setting 13.6 s

Times are from the question being sent to the finished answer. The quickest mode for each question is in bold.

Method and limits

- 11 questions from 10 conversations, including follow-up questions. Each mode answered each question once.
- All modes used the same main model (gemini-3.7-flash), the same warehouse data and the same answer checks. Only the way the prompt is built differs.
- The tests ran on a separate copy of the server, so no real users or shared data were involved.
- Answers were judged blind by two AI judges. AI judges can be wrong, and they do not always agree.
- This is a small sample. A different set of questions could change the order, especially where the gaps are small.
- Gemini's speed changes from run to run and hour to hour, so treat time differences under a couple of seconds as noise.
- Vanilla and EXP-RAG answers, and the Multi-Agent answers to simple questions, come from one test session (round 3). Multi-Agent team answers and the fast setting come from a session about an hour later (round 4), so speed carries some session-to-session noise.
- Round 3's Multi-Agent team answers used an earlier low-thinking setup and are not in the headline.
- Total Gemini spend for all evaluation rounds, including warm-ups and smoke tests, was about $0.42.

Source data: docs/reports/data/multi-agent/final (answers.json, key.json, judge-a.json, judge-b.json).